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Work Experience

May 2024 to present **AI Developer**, *Nokia Co.*, Ottawa, Canada

- AI Automation project to create a Jira ticket (project) and a project status report on a confluence page:
 - Multi-agent Orchestra: LangGraph and NeMo AgentIQ.
 - MCP (Model Context Protocol) server Deployment on ACI (Azure Container Instance).
 - Automation Frameworks: N8N and Azure Logic App.
- Build a chatbot project, deploying a RAG system for PDF Legal Documents (summarize and classify paragraphs):
 - Frameworks: Llama-Index, LangChain, LangGraph, CrewAI.
 - Deploying Frameworks: vLLM, Ollama, TGI and TensorRT-LLM (Triton-LM)
 - LLM: Llama 3.1, Qwen, Llava, and Gemini 2.0
 - GraphRAG: Neo4j and Microsoft GraphRAG
 - VectorDB: Weaviate, CosmosDB, Faiss, and Pinecone
 - Back/Front end: FastAPI and React
 - Hallucination: CRAG-Agent (Self-Corrective RAG)
 - Fine-Tuning LLM Model: Lora and Peft.
 - Retrieval Evaluation: Recall@K and MRR (Mean Reciprocal Rank)
 - Generation Evaluation: ROUGE and BLEU.
- Changing logos in videos/images project:
 - Tools: Stable Diffusion Automatic1111 WebUI and ConfyUI.
 - Object Detection Frameworks: Tensorflow Object Detection and MMDetection.
 - Object Detection Models: Fine-tuning Faster R-CNN, RTMDet, Swin Transformer, DETR, ViTDet, MViTv2, Deformable DETR, DINO, and MViTv2 (ViT)
 - Fine-tuning Stable Diffusion (Inpaint) and Object Detection Models.
 - Back/Front end: Flask and HTML.

Jan. 2020 to Jan. 2024 **Data Scientist**, *BI Expertise Co.*, Quebec City, Canada

- Drone Position (and Orientation Estimation) project:
 - Data sets: IMU (accelerometer, gyroscope) and GPS sensor data.
 - Sensor Fusion: Kalman Filter (KF), Extended Kalman Filter (EKF), Particle Filtering (PF), Unscented KF(UKF) and the Cubature KF.
 - Designed and coded the system to estimate 3D position and velocity, optimizing noise reduction and accuracy compared to raw sensor inputs.

- Programming Languages: C++ and Python.
- Pedestrian and vehicle tracking with Drone project:
 - Optimize object detection models for On-Device deployment: Converting model to TensorFlow Lite, Quantization, Pruning, and Distillation.
 - MOT(multi-object tracking) Trackers: ByteTrack, BoT-SORT, OC-SORT, DeepSORT, and JPTrack.
 - Adding a lightweight EKF (and CKF) layer (from IMU/GPS) to lift 2D tracks to 3D—boosts to predict 3D states (position/velocity) accounting for drone yaw/acceleration. Using trained object detection models: YOLO-N, YOLOv8n, YOLOv5n, YOLO-NAS small architectures.
 - Select hardware: Combination of Raspberry Pi, ARM-based CPUs, TensorFlow Lite C++ API, DJI SDK was used for first version of Drone then it was upgraded to NVIDIA Jetson Platform + TensorFlow Lite C++ API with GPU Delegate + DJI SDK.
- 3D Object Detection project on Videos to Detect Vehicles and Pedestrians:
 - MMDetection3D Models: DETR3D, Swin3D, and YOLOv5/4.
 - Programming Languages: C++ and Python.
 - Point-Cloud Data sets: Point-Pillars, Second, and CenterPoint.
- Build a Chatbot project to extract information from documents (images/PDF) with LLM:
 - Frameworks: LangChain, LangGraph, NeMo AgentIQ,
 - MM-LLM (Multi-Modal LLM): Llama 3, Qwen 2.5, and Llava-v1.6
 - OCR Tools: Tesseract and Amazon Textract.
 - Back/Front end: FastAPI and React
 - Hallucination: Retrieval-Augmented Verification (RAV).

Jan. 2019 to **Software Engineer**, *Laval University*, Quebec City, Canada

- Jan. 2020
- Finding Positive/Negative Comments Project (Sentiment Analysis of NLP).
 - Sentiment analysis with NLTK and Spacy.

Jan. 2017 to **Data Engineer**, *Promotions VPS Co.*, Quebec City, Canada

- Jan. 2019
- Data visualization with Tableau and Power BI
 - Extract Transform Load (ETL) with SSIS, SSAS, SSRS by writing SQL query on Microsoft SQL and MySQL

Technical Skills

Programming Python, R, Java, HTML, JavaScript, Matlab and C++

SQL Microsoft SQL, MySQL, Oracle, BigQuery, and ETL (Extract Transform Load) with SSIS, SSAS, SSRS.

Education

Certification **Azure Cloud certification (DP-100) from Microsoft**

Sept. 2014 to **Master of Science**, *Engineering*

Jan. 2017 Laval University, Quebec City, Canada